



PAPER A



2018 SCIENCE

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

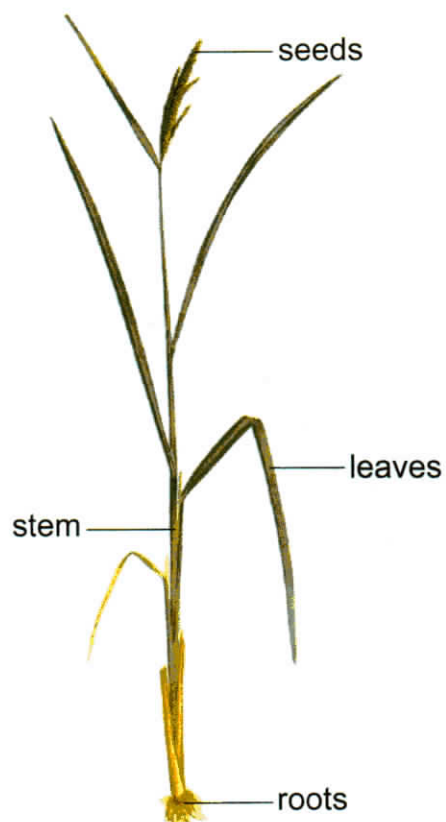
1. This photo shows some grains of rice.



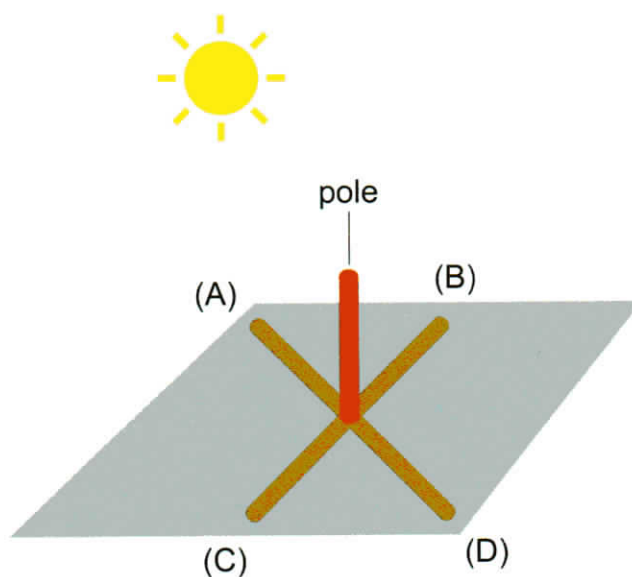
The diagram on the right shows a rice plant.

Which part of the plant has the grains of rice?

- (A) roots
- (B) leaves
- (C) stem
- (D) seeds

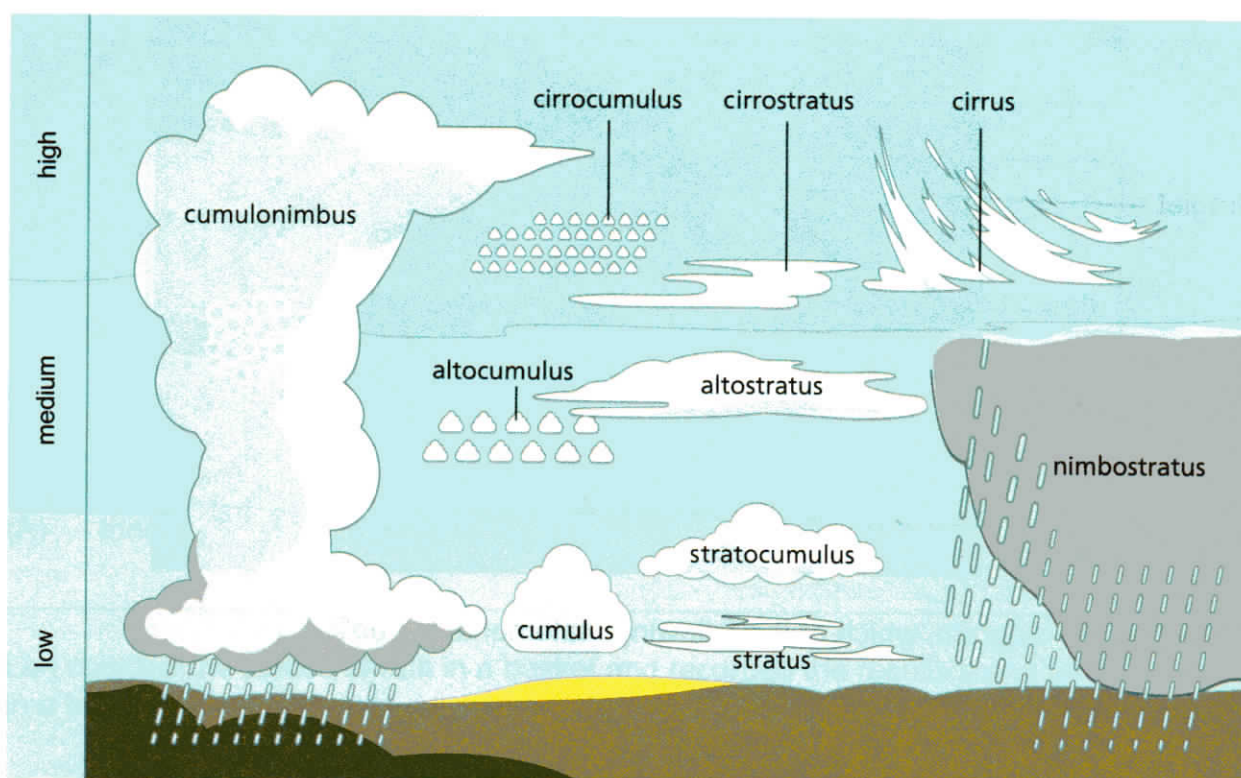


2. Which option correctly shows the shadow cast by the pole?



For questions 3 and 4 use the information below.

The diagram shows the names and heights of different types of clouds.



3. How many types of clouds are shown in the diagram?

- (A) 2
- (B) 3
- (C) 10
- (D) 13

4. Based on the diagram, which clouds bring rain?

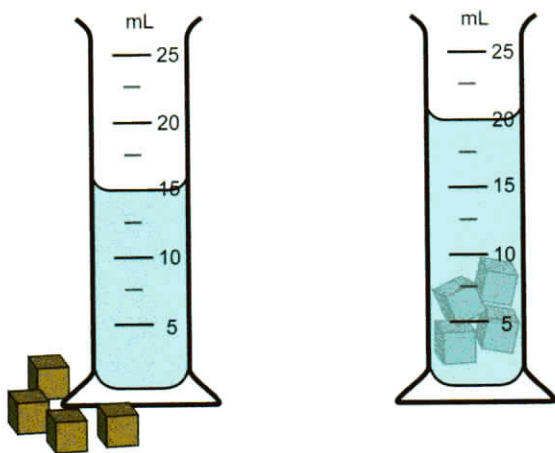
- (A) nimbostratus and cumulonimbus
- (B) cumulonimbus and cirrocumulus
- (C) cirrocumulus and altocumulus
- (D) cumulus and nimbostratus

5. This diagram shows the life cycle of a frog.



Based on the diagram, which of the following statements is true?

- (A) Tadpoles are born with legs.
 - (B) Frogs can survive out of water.
 - (C) Frog eggs are attached to plants on dry land.
 - (D) Frogs remain the same colour throughout their life cycle.
6. When you place an object under water, it pushes the water out of the way and the water level goes up. This is called displacement.



Jane placed 4 cubes in 15 millilitres (mL) of water.

How much water, in mL, did the cubes displace?

- (A) 4
- (B) 5
- (C) 15
- (D) 20

7. Animals use their senses to explore their surroundings. The table shows the five senses people have and what they sense.

Sense	Sense organ	What is sensed
hearing	ear	vibrations in the air
sight	eye	light
smell	nose	chemicals in the air
taste	tongue	chemicals on objects
touch	skin	pain, pressure, temperature and touch

Butterflies use their feet to sense the chemicals on the plant they are standing on.

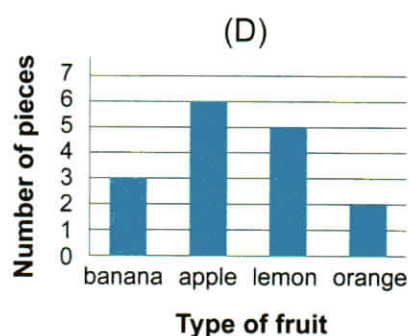
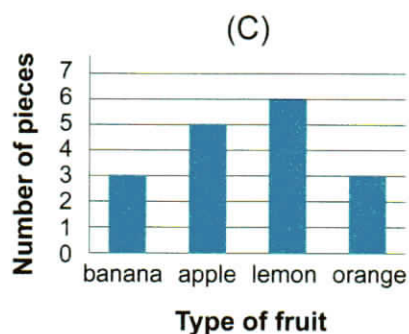
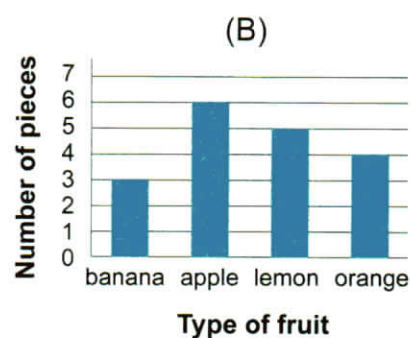
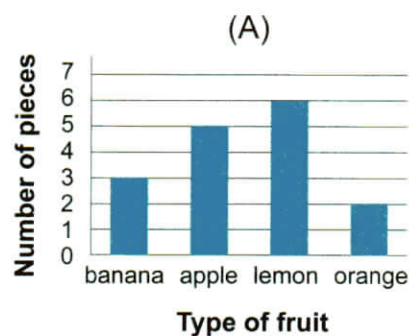
Which sense organ of people has a similar use to that of a butterfly's feet?

- (A) tongue (B) skin (C) nose (D) feet

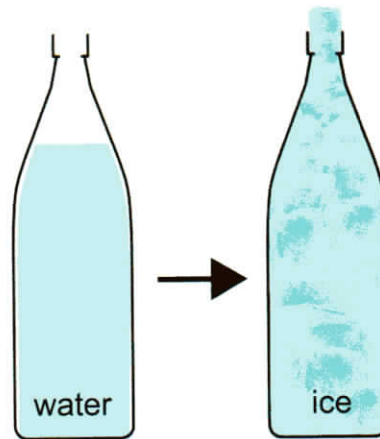
8. Lan counted the pieces of fruit in a basket and recorded the results using tallies, in a table.

Type of fruit	Number of pieces
banana	III
apple	### I
lemon	###
orange	II

Which graph correctly represents the information in the table?



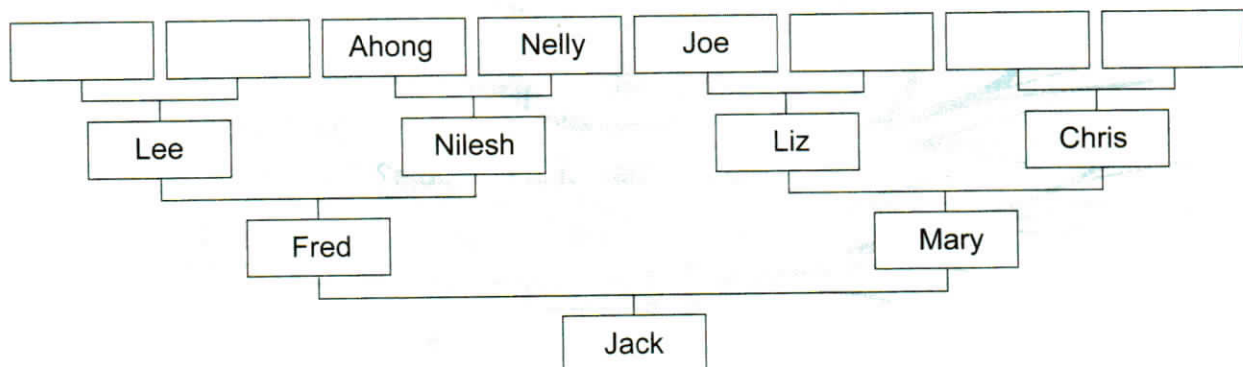
9. Jason placed a bottle of water in the freezer. When he took the bottle of water out of the freezer the next morning, this is what he saw.



What is the best explanation for this?

- (A) Water takes up more space when frozen.
- (B) Water becomes heavier when frozen.
- (C) Water bubbles up when freezing.
- (D) Water spills before freezing.

10. Jack is drawing his family tree. The diagram shows what he has already drawn.



Liz and Chris are two of Jack's grandparents.

Who are two of Fred's grandparents?

- (A) Lee and Niles
- (B) Nelly and Joe
- (C) Ahong and Nelly
- (D) Liz and Chris

11. Jay's mum is making some dough. She has some flour in a bowl. She tells Jay that she will need $1\frac{1}{2}$ cups of water.



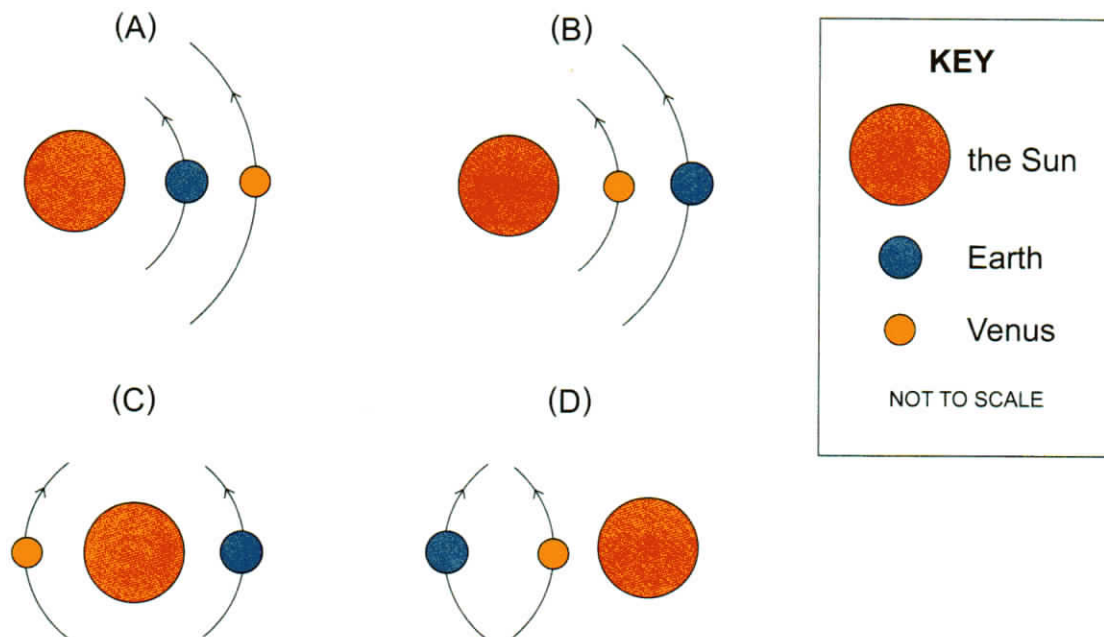
Jay's mum then decides to double the amount of flour in the bowl to make more dough.

How many cups of water will she need altogether to make the dough?

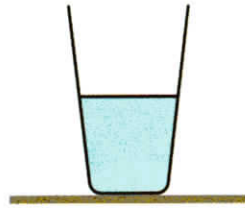
- (A) $1\frac{1}{2}$ (B) 2 (C) $2\frac{1}{2}$ (D) 3

12. The transit of Venus occurs when the planet Venus travels between the Sun and Earth. It can be observed from Earth.

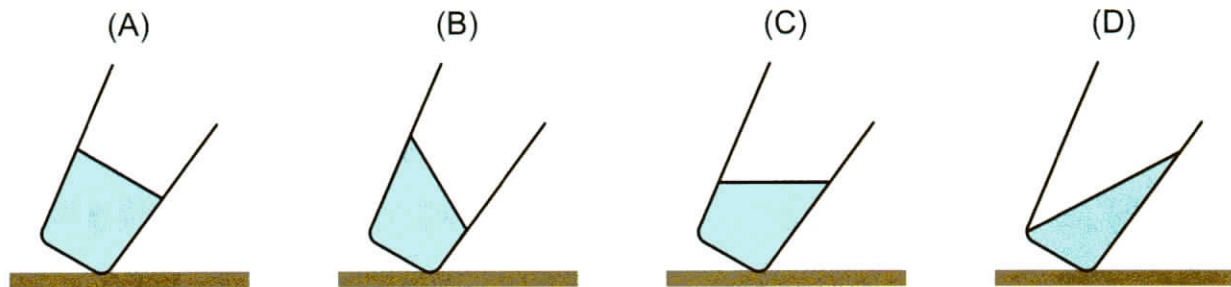
Which diagram correctly shows the position of the Sun, Venus and Earth when a transit of Venus takes place?



13. The picture shows a glass with some water in it.



Which picture correctly shows the water level in the glass when it is tilted?



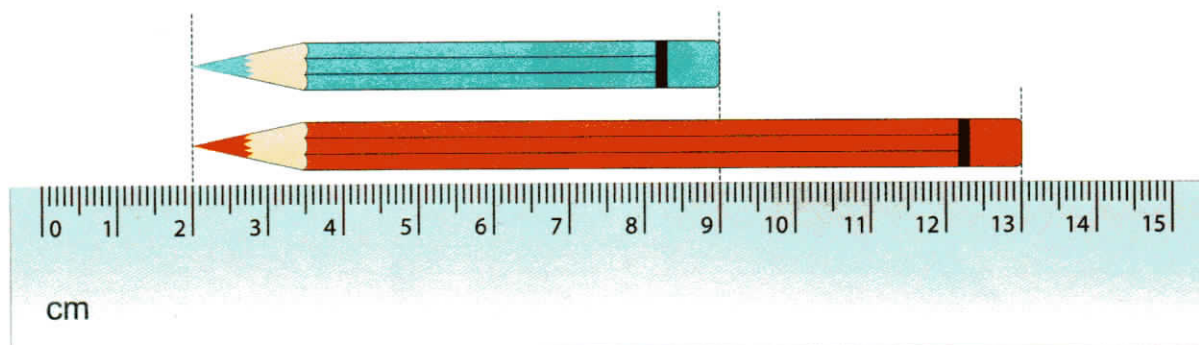
14. Nutrition labels on food packaging show the amount of the food a person should have at one time. This amount is called a 'serving size'.

The diagram shows part of the label on a packet of chocolate.

Nutrition Information	
Servings per pack:	5
Serving size:	10 grams

How many grams of chocolate are there in the packet?

15. Dan has two coloured pencils.



What is the difference in length between the two coloured pencils?

- (A) 2 cm (B) 4 cm (C) 6 cm (D) 7 cm

16. To conduct a fair test, the investigators must:

- **change** only one thing for each test
- **keep** everything else the same, and
- **observe or measure** one change that happens.

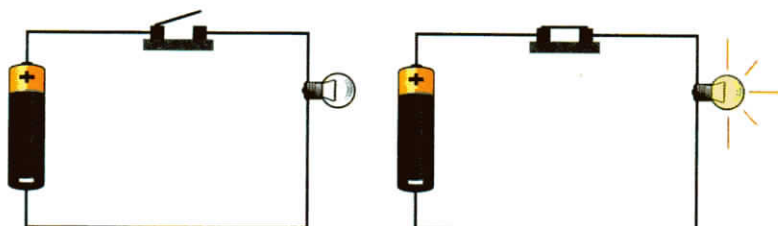
A group of Year 3 students want to see if the shadow of a stick changes length during the day.

What must the students change, what must they keep the same, and what must they measure to make their investigation a fair test?

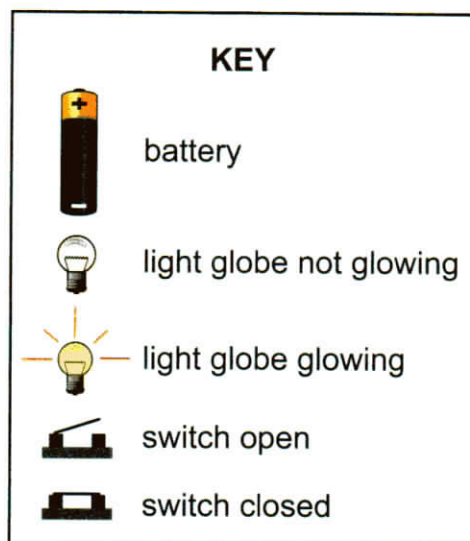
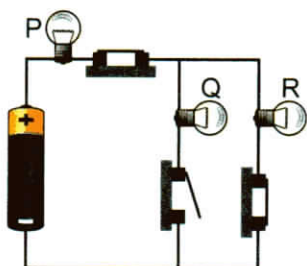
	Change	Keep the same	Measure
(A)	the time of the day	the length of the shadow	the length of the stick
(B)	the length of the shadow	the length of the stick	the time of the day
(C)	the time of the day	the length of the stick	the length of the shadow
(D)	the length of the stick	the time of the day	the length of the shadow

17. Sally is investigating electrical circuits.

She uses some diagrams to help her work out what is needed to make a light globe glow.



Sally wires up this circuit. The light globes all work and the battery produces enough electricity to make them all glow.

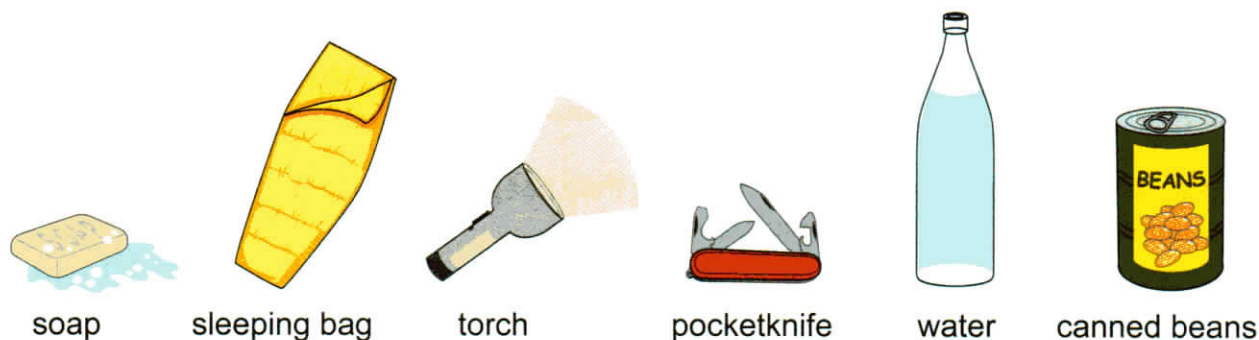


Which light globes in her circuit will glow?

- (A) P and R only (B) P and Q only (C) Q and R only (D) P, Q and R

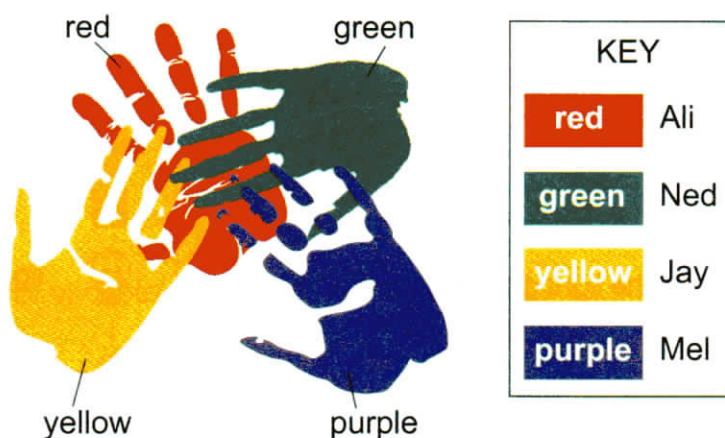
18. People need to breathe air, drink water, eat food and keep warm to stay alive.

Jen is going camping in the mountains for three days. She is preparing things to take with her from these:



Which of the following are the three most important things for her to take?

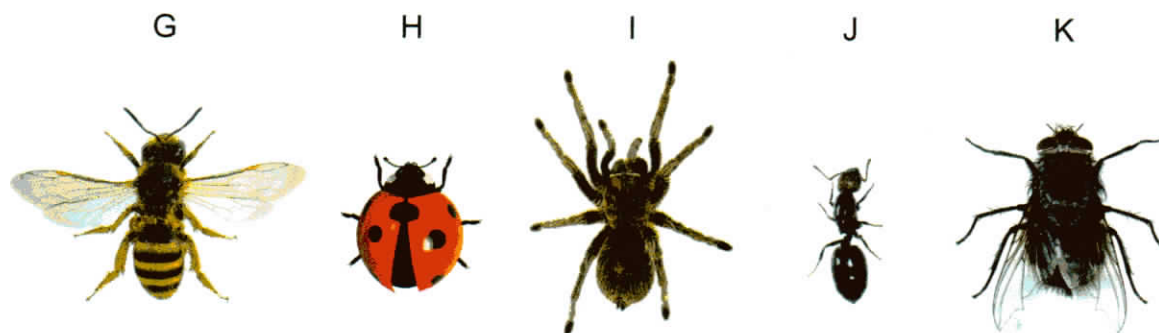
- (A) sleeping bag, torch, water
(B) torch, pocketknife, sleeping bag
(C) soap, sleeping bag, water
(D) sleeping bag, water, canned beans
19. Ali, Ned, Jay and Mel made an artwork using palm prints. They dipped one hand into the paint and then pressed their palms onto a piece of paper one after another.



In which order did they press their hands on the paper?

	1st	2nd	3rd	4th
(A)	Ali	Ned	Jay	Mel
(B)	Mel	Ned	Jay	Ali
(C)	Ali	Jay	Ned	Mel
(D)	Jay	Ali	Mel	Ned

20. Brian is looking at the photographs of some animals. All the animals have antennae or pedipalps on their heads.



He places the animals into two groups, based on one particular feature.

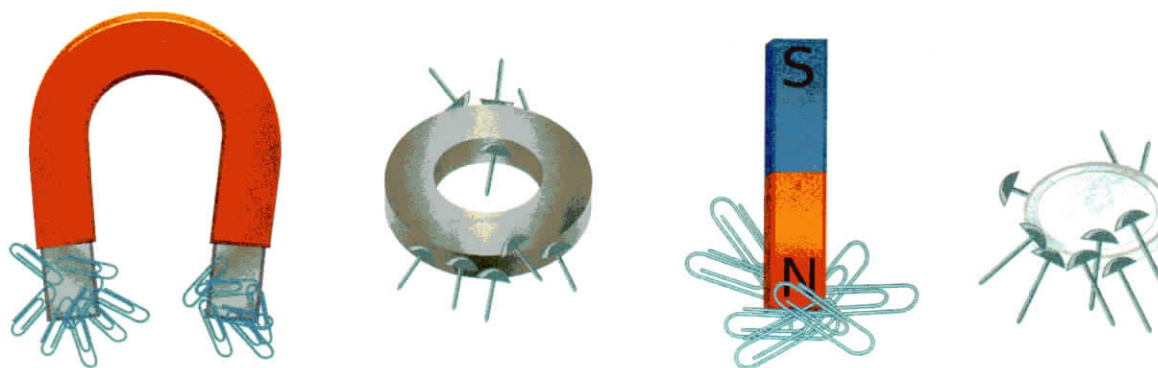
Group 1: G, H, K Group 2: I, J

What feature does he use to decide which group to place them in?

	Group 1	Group 2
(A)	spots	no spots
(B)	six legs	eight legs
(C)	a large body	a small body
(D)	wings	no wings

21. Magnets can attract objects made of iron.

Ying wants to compare the strengths of four magnets. She used some paper clips and thumb tacks to carry out her investigation. This shows her results.

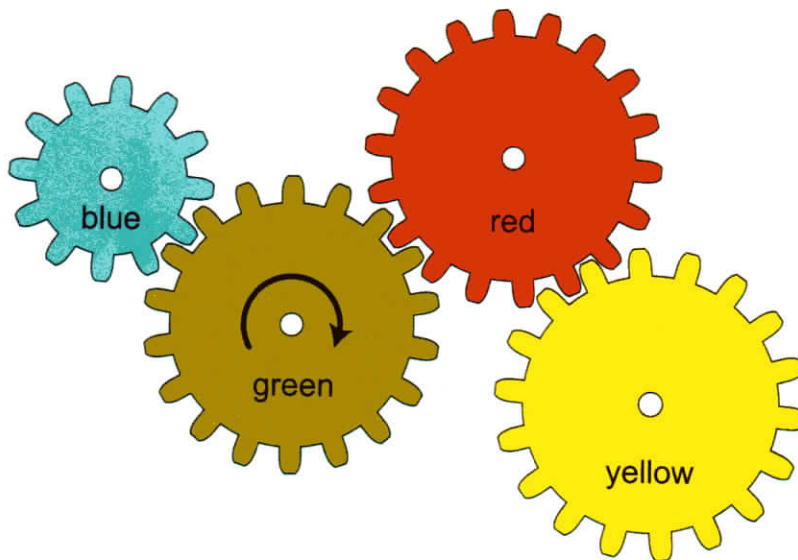


This investigation is not a fair test.

What can Ying do to make it a fair test?

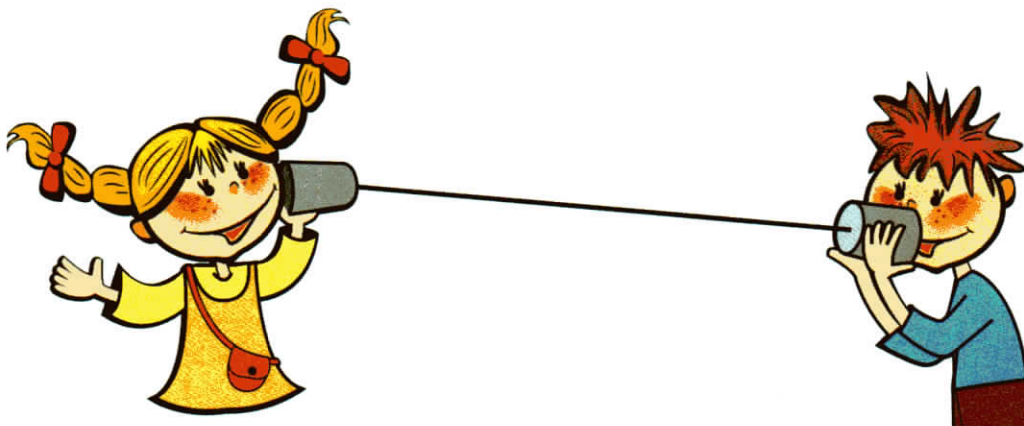
- (A) use fewer paper clips
- (B) use the same size paper clips for each magnet
- (C) use the same number of paper clips and thumb tacks for each magnet
- (D) use large paper clips for the large magnet and small paper clips for the small magnet

22. The diagram shows three gears. Each gear has teeth that link to another gear. The green gear is turned in the direction shown by the arrow.



Which gear will turn the most number of times in one minute?

- (A) red
(B) green
(C) blue
(D) yellow
23. Sound can travel through different materials. When we speak to each other, the sound travels through the air to reach our ears.
- Jill and David want to find out whether the material that a sound travels through has an effect on how it sounds. They made a string telephone like this one, using a cotton string and two cups.

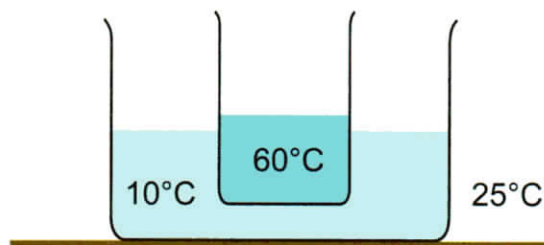


What else do they need for their investigation?

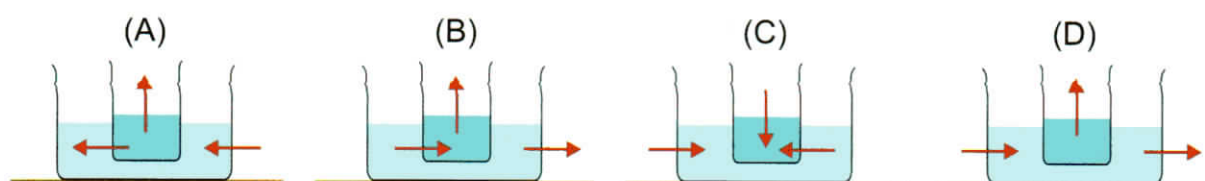
- (A) a real telephone
(B) strings made of other materials
(C) cotton strings of different lengths
(D) other people to listen to the telephone

24. When two objects are in contact with one another, heat flows from the object with the higher temperature to the object with the lower temperature.

The diagram shows a small beaker of water placed in a larger container of water on a table. It also shows the temperature of the water in each container and the air temperature.



Which diagram correctly shows the direction in which heat flows at this time?



25. This table lists the time of sunrise in Sydney over seven days in May 2018.

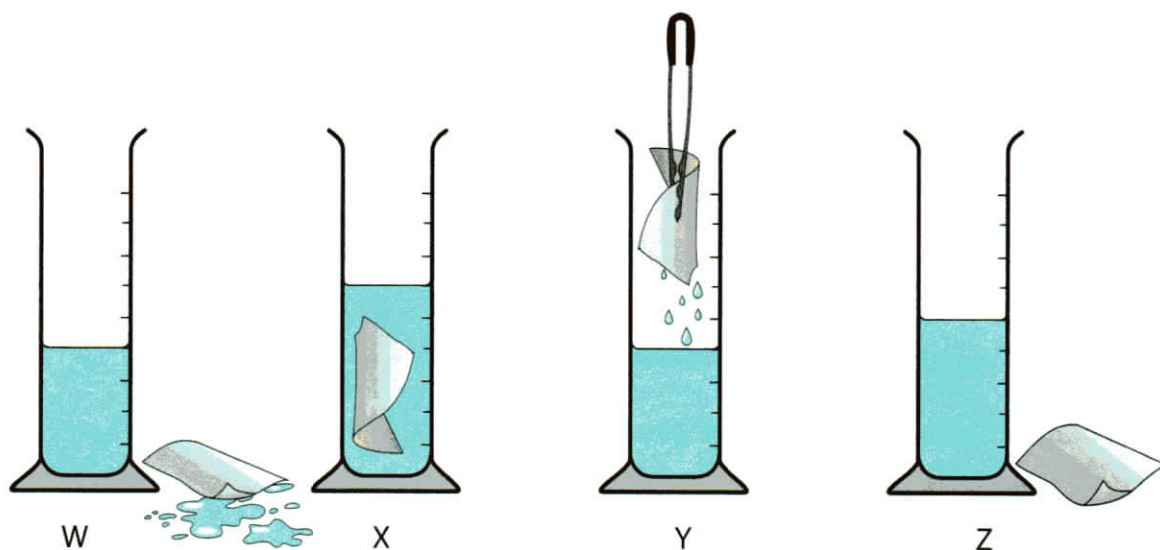
Date	Sunrise (am)
May 23	6:45
May 24	6:46
May 25	6:46
May 26	6:47
May 27	6:48
May 28	6:48
May 29	6:49

Based on the information in the table, what is the most likely time of sunrise on May 31, 2018?

- (A) 6:49 am
(B) 6:50 am
(C) 6:51 am
(D) 6:52 am

26. Laura carried out an investigation to find out how much water a paper towel could absorb.

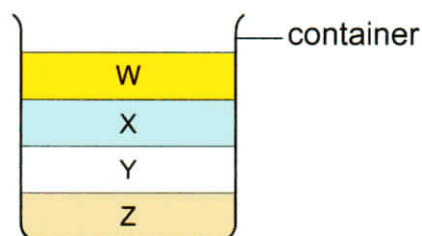
The diagram shows four steps of her investigation, but not in the correct order.



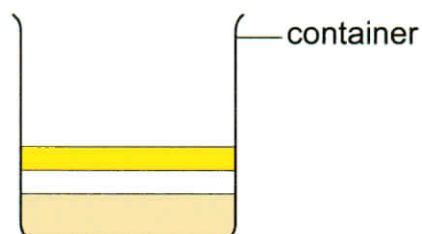
Which of these lists the steps in the correct order?

- (A) W, Y, X, Z (B) Z, W, X, Y (C) Y, X, W, Z (D) Z, X, Y, W

27. The diagram shows four layers of different liquids in a container. The liquids do not mix with each other.



There is a small leak in the wall of the container. After some time, the leak was plugged. The diagram shows what the liquids looked like after the leak was plugged.



From which layer in the container did the liquids leak?

- (A) W (B) X (C) Y (D) Z

28. Seeds of different plants vary in the time they take to sprout at different temperatures.

The table compares the sprouting time of the seeds of four different plants.

Days it takes to sprout at different temperatures

	10 °C	15 °C	20 °C
onion seeds	13	7	5
spinach seeds	12	7	6
lettuce seeds	7	4	3
cabbage seeds	17	10	7

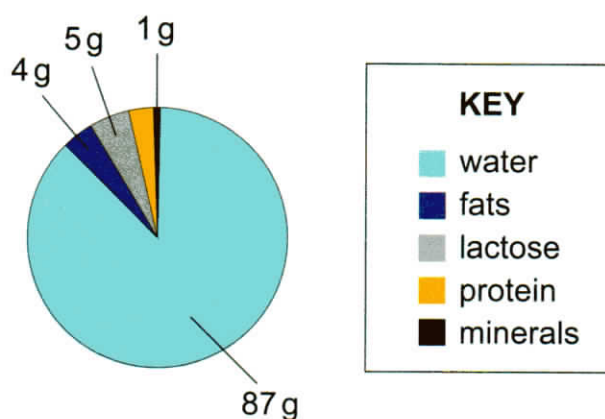
Which seeds will sprout first at any temperature?

- (A) onion
- (B) spinach
- (C) lettuce
- (D) cabbage

For questions 29 and 30 use the information below.

Milk is made up of water and some solids. The solids include fats and non-fat solids.

The sector graph shows what is in 100 grams (g) of cow's milk.



29. Which part of the milk has not been labelled with its amount?

- (A) fats
- (B) lactose
- (C) protein
- (D) minerals

30. Based on the diagram, how many grams of non-fat solids are there in 100 g of cow's milk?

- (A) 3
- (B) 9
- (C) 10
- (D) 13

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SOURCES

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THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

Australia	Year 3
Brunei	Primary 3
Egypt	Year 3
Hong Kong	Primary 3
Indian Subcontinent¹	Class 3
Indonesia	Year 4
Malaysia	Standard 3
Middle East²	Class 3
New Zealand/Pacific³	Year 4
Singapore	Primary 2
Southern Africa⁴	Grade 3

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria, Jordan and Pakistan.

³ Pacific Region: Vanuatu, Papua New Guinea and Fiji.

⁴ Southern Africa Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.

PAPER

A



UNSW Global



Question Number	Area Assessed	Description	Correct Answer
1	Observing/Measuring	Recognise observable properties of a living thing	D
2	Predicting/Concluding	Predict a simple cause and effect	D
3	Observing/Measuring	Recognise different types of clouds in a diagram	C
4	Interpreting	Interpret a diagram	A
5	Interpreting	Interpret a diagram showing the life cycle of a frog	B
6	Reasoning/Problem solving	Determine the volume displaced by some solids	B
7	Predicting/Concluding	Draw a conclusion from tabulated data	A
8	Interpreting	Determine which graph matches tabulated data	D
9	Reasoning/Problem solving	Use reasoning to infer the cause of a phenomenon	A
10	Interpreting	Interpret relationships in a diagram of a family tree	C
11	Observing/Measuring	Calculate a total volume in given scenario	D
12	Interpreting	Interpret a diagram illustrating an astronomical phenomenon	B
13	Predicting/Concluding	Predict a simple effect caused by a given change	C
14	Predicting/Concluding	Calculate a value based on nutritional information	A
15	Observing/Measuring	Measure the difference in the lengths of two objects	B
16	Investigating	Identify three types of variables in a simple investigation	C
17	Reasoning/Problem solving	Solve a problem about an electrical circuit	A
18	Interpreting	Identify the items essential for a given excursion	D
19	Reasoning/Problem solving	Determine the order of a series of actions based on the final product	C
20	Predicting/Concluding	Determine the criterion by which given animals have been grouped	D
21	Investigating	Recognise the conditions for a fair test in a given investigation	B
22	Reasoning/Problem solving	Solve a simple problem in a gear system	C
23	Investigating	Identify the equipment needed for an investigation	B
24	Predicting/Concluding	Predict the result of a given scenario	A
25	Predicting/Concluding	Draw a conclusion based on data given in a table	B
26	Investigating	Sequence the steps of an investigation using diagrams	D
27	Reasoning/Problem solving	Use reasoning to infer the cause of a phenomenon	C
28	Interpreting	Interpret information in a table	C
29	Interpreting	Interpret a sector graph	C
30	Reasoning/Problem solving	Determine a value using information in a sector graph	B